

4416 Vincent Ave. S.
Minneapolis, MN 55410

James Jagielski
jj3552@columbia.edu
612-306-2772

<https://www.linkedin.com/in/james-jagielski/>

EDUCATION

- Columbia University** | *Candidate in MS/PhD under Ken Sheppard* **New York, NY**
• **Clubs:** Sailing and Table Tennis. **Relevant Coursework:** VLSI Design Lab, Advanced Analog IC, mmWave IC, microwave circuit design, Analog-Digital Interfaces, Power Management IC, Harsh Environment electronics, Fall 2026
- Olin College of Engineering** **Needham, MA**
• Bachelor of Science degree in Electrical and Computer Engineering May 2025
• Relevant Coursework: Mixed Analog-Digital VLSI, Introduction to Microelectronic Circuits, Software Design, GPA: 3.85/4.0
Computer Architecture, Advanced Computer Architecture, Power Electronics, Sensors & Instrumentation
- Southwest High School** **Minneapolis, MN**
• IB Diploma; Varsity team captain of swimming, sailing and table tennis June 2021 | GPA: 3.95/4.0

EXPERIENCE

- Columbia University** **New York, NY**
Bioelectronic Systems Laboratory | Ken Shepard Spring 2026
• Designed a Switched-Cap fully differential amplifier for 5 μ W with 76dB open loop gain in a sensing and stimulation device for dry eye disease, placed in the tear duct. This circuit will be taped out in TSMC 130nm and will be tested Fall 2026.
• A 1 MHz Buck Converter with Type-III Compensation. Converted 3V to 1V. Focused on the Differential Difference Amplifier.
• Closed-Loop Dynamic Biasing for Threshold Voltage Stability in LEO Beamforming Systems.
• 60 GHz receiver in 90nm with LNA, MIXER, and VCO. I was focused on the VCO and full integration.
Course Assistant | Digital VLSI Circuits Fall 2025
• Supported students in developing digital VLSI design techniques in TSMC 65nm and finFET N16 TSMC technology.
Course Assistant | Digital Systems Laboratory Spring 2026
• Basic lab course in the design of digital systems. Support students in legacy LSI logic families (such as TTL) as well as data conversion, timing circuits, data communications, and programmable logic on bread board circuitry.
- Semtech Corporation** **San Diego, CA**
Test Engineering Intern Summer 2024
• Wrote test programs for post-silicon testing on Advantest V93000 and Teradyne flex
• Worked on testing communication protocols for various chip packages
• Simulated and designed electromagnets in COMSOL for a magnetic sensor interposer board
- Olin College** **Needham, MA**
MADVLSI Circuits & Systems Lab | Dr. Bradley Minch Summer and Fall 2023
• Designed a source measuring unit on a 4-layer PCB (to characterize semiconductor devices) for use in the microelectronics course at Olin, wrote the associated firmware, and made a MATLAB GUI for easier use
Olin Plasma Engineering Lab | Electrical Sub Team Lead Fall 2023
• Worked on the high-power circuitry for the hall thruster and set up electrical circuitry in plasma labs at MIT and Northeastern
• Designed a testing rig for the hall thruster to measure the force and temperature and collected the data for physics analysis
• Designing a microwave frequency electrical system for an Electron Cyclotron Resonance Thruster
Course Assistant | Introduction to Microelectronic Circuits Spring 2024
• Supported students in developing an intuition and methodologies for transistor-level design and operation of single-transistor amplifier stages, current mirrors, differential pairs, and single-stage op-amps
Course Assistant | Introduction to Sensors, Measurement, and Instrumentation Spring 2023
• Counseled students in designing analog signal processing circuits with an emphasis on op-amps, and passive and active filters

SKILLS

- **Electrical Design and Prototyping:** Cadence Virtuoso, ADS Momentum, Magic (VLSI Layout), DRC/LVS, postlayout simulations, xschem, NGSpice, LTSpice, KiCAD, Altium, Monte Carlo Simulations, floorplanning, power analysis, layout versus schematic, block diagrams, design reviews
- **Electrical Instrumentation:** Oscilloscope, Function Generator, Source Measure Unit, Multimeter, Current Clamp
- **Software and programming:** MATLAB, Python, C, Verilog, GTKwave, COMSOL, SOLIDWORKS
- **IC Circuit Design Experience:** Bandgap reference, op-amps, comparators, biasing circuitry, analog-to-digital converters, digital-to-analog converters, current mirrors, switched-capacitor circuits, digital logic cells, low-power moderate inversion CMOS design

PROJECTS

A Bit-Serial Algorithmic Bi-Directional A/D/A Converter in SKY130 Process

Mixed Analog-Digital VLSI

- Designed 8-bit A/D/A converter and explored floating gate differential pairs for the OTAs and comparator

Self-Biased Two-Stage Cascode Op-Amp in SKY130 Process

Mixed Analog-Digital VLSI

- Converted an op-amp design from 5V process to a 1.8V process using low voltage current mirrors
- Optimized matching using proper analog layout techniques (common centroid, same surround, and dummy devices)

7-bit M-2M Digital-to-Analog Converter in SKY130 Process

Mixed Analog-Digital VLSI

- Designed a current bias for the D/A converter using a bootstrap current mirror
- Designed a CSRL Shift Register to insert bits into the D/A converter
- Simulated the differential and integral nonlinearity and Monte Carlo to examine accuracy of the D/A converter

RISC-V Single-Cycle Scalar CPU

Computer Architecture

- Implemented CPU using structural Verilog, wrote test benches and simulated using GTKwave

Asynchronous SAR ADC in SKY130 Process

Mixed Analog-Digital VLSI

- Designed an ultra-low power level crossing SAR ADC with handshake based digital control circuitry

MD5 Hashing Accelerator TapeOut in SKY130 Process

AVLSI LAB

- Designed, laid out, and tested custom standard cells for the Asynchronous VLSI research lab at Yale University